From rows to claims

Coding Lesson 3 · Math Camp 2026

Rony Rodriguez-Ramirez

Harvard Kennedy School · API 209

Monday, August 24, 2026

Lesson map

Capabilities for a changing labor market

Define the policy task

Compare economies and changes over time

The specification determines the variation used to estimate the coefficient

Check one briefing sentence with Codex

From lesson to lab

Capabilities for policy analysis in a changing labor market

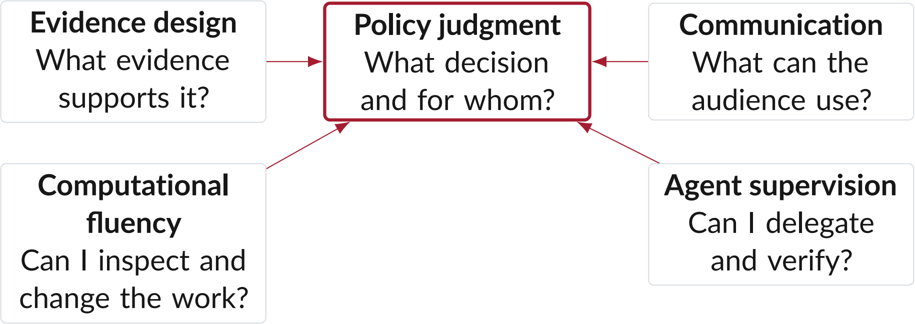
Employers expect roughly 39% of workers' core skills to change by 2030. Analytical thinking remains the most frequently reported core skill, while AI, big data, and digital skills are among the fastest-growing areas.

OECD vacancy evidence also shows strong demand in AI-exposed occupations for:

- management and business-process skills;
- cognitive, social, and communication skills;
- digital fluency across changing tools.

Coding agents draft code, inspect files, and run checks. Analysts remain responsible for the policy question, empirical design, and recommendation (World Economic Forum, 2025; Green, 2024; OECD, 2026).

Five capabilities used in the policy briefing



The Lesson 3 briefing uses all five capabilities.

Four terms for the agent workflow

Model
Parameters that receive context and generate tokens. Model choice affects capability, speed, and cost.

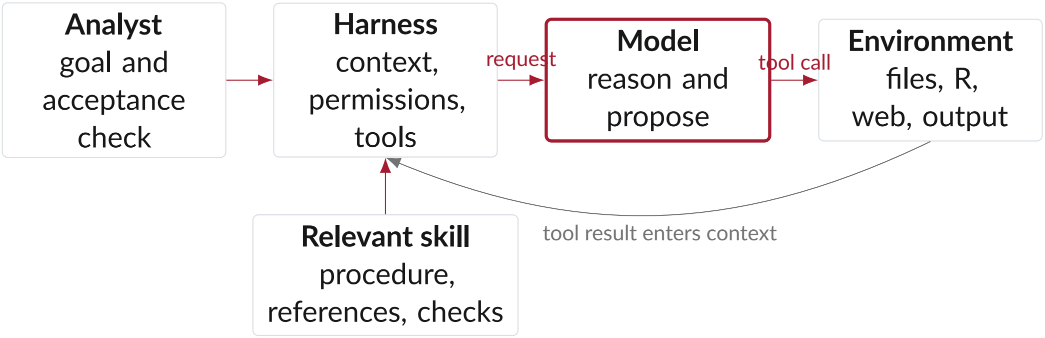
Reasoning
Additional model computation before an answer or tool request. Higher effort can help with difficult comparisons and consumes more output tokens.

Agent
A model operating through a harness with context, tools, permissions, and a continuing session. Codex is an agent.

Skill
Reusable task instructions and resources loaded when relevant. A skill supplies a procedure and checks; model parameters remain unchanged.

Adapted from Pocock's AI Coding Dictionary entries on models, agents, harnesses, tools, effort, and skills (Pocock, 2026).

One agent turn can contain several model requests



The model proposes tool calls. The harness executes them and returns the results. The analyst reviews the evidence and decides whether the task is complete.

Adapted from Pocock's descriptions of model-provider requests, tools, tool results, harnesses, and skills (Pocock, 2026).

Analogy

A useful analogy:

- Model = which person you hire.
- Reasoning ability = how capable that person is.
- Effort = how long and carefully they work.
- Speed = how soon you receive the result.

Rank the capabilities for a cabinet briefing

Slido ranking

A coding agent produced a figure and paragraph for a ministerial briefing due in 30 minutes. Rank all five capabilities. Place the two that most constrain whether you can use the work first.

1. Frame a policy question with a decision at stake.
2. Exercise statistical judgment.
3. Steward data and provenance.
4. Read, modify, and explain code.
5. Supervise and verify an agent.

Join at sli.do
#2209653

Route for the lesson

Minutes	Phase	Evidence
00–20	Question + panel	Agent workflow, rank, unit, key, period, and row count
20–47	Figures	Pooled plot, trajectories, and Checkpoint 1
47–57	Break	Ten minutes away from the screen
57–91	Models	Three specifications and Checkpoint 2
91–116	Claim check	One briefing sentence checked against the model results
116–120	Lab handoff	Closing rank and analysis sequence

The policy director's question

Briefing request

Across 2000–2022, how did national income and under-five mortality move together across economies and within economies? Which associations does this evidence support in a briefing to a policy director?

The analysis must distinguish:

- an **economy–year** from an economy;
- differences **between economies** from changes **within economies**;
- a descriptive association from an intervention effect;
- a pooled association from a within-economy association.

The policy question determines the relevant comparison.

Recall the Lesson 2 analysis table

Before seeing any output, write three answers:

1. What does one row represent?
2. Which columns form the unique key?
3. Which two checks must pass before we draw a figure?

Verify the panel in RStudio

Before the code runs, predict whether the duplicate-key result should equal zero. State what a nonzero result would mean for the analysis.

```
panel_record <- health_panel |>
  summarise(
    rows = n(),
    economies = n_distinct(iso3c),
    years = n_distinct(year),
    duplicate_economy_years =
      anyDuplicated(data.frame(iso3c, year))
  )
```

Record the unit, the key, the period, and the duplicate-key result.

The verified panel

4,296 economy–years

188 economies · 23 years · 2000–2022

The key `iso3c + year` is unique. Coverage ranges from 185 to 188 economies per year. Every retained row has both income and mortality observed.

The panel supports cross-economy comparisons within a year and changes within the same economy over time.

Lesson map

Capabilities for a changing labor market

Compare economies and changes over time

Predict, plot, interpret

The specification determines the variation used to estimate the coefficient

Check one briefing sentence with Codex

From lesson to lab

The policy question contains two comparisons

1. Do lower-income and higher-income economy–years occupy different parts of the mortality distribution?
2. As the same economy changes over time, does its trajectory move in the same direction as the pooled association?

The pooled scatterplot addresses the first comparison. Economy trajectories display the second.

State the comparison before choosing the graph.

Predict the pooled relationship

Slido multiple choice

We will plot all 4,296 economy–years and use logarithmic scales on both axes. Which pattern do you expect?

1. A downward band with substantial dispersion.
2. A downward band with nearly identical paths across economies.
3. A flat cloud after the logarithmic transformations.
4. Twenty-three separate bands, one for each year.

Record your prediction. We will compare it with the plot.

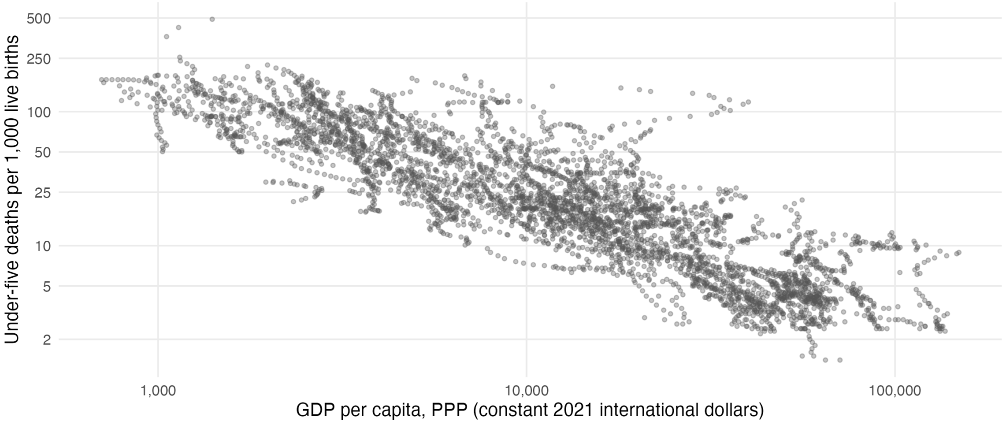
Join at sli.do

#2209653

Build the pooled plot in RStudio

```
ggplot(
  health_panel,
  aes(
    x = gdp_per_capita_ppp,
    y = under5_mortality
  )
) +
  geom_point(alpha = 0.34) +
  scale_x_log10() +
  scale_y_log10()
```

Higher income is associated with lower under-five mortality



Source: World Bank, World Development Indicators. Each point is one observed economy–year from 2000–2022. Both axes use logarithmic scales.

Read the pooled plot

Direction	Higher GDP per capita is associated with lower under-five mortality.
Dispersion	Economies at similar income levels can have substantially different mortality.
Departure	Some observations have much higher mortality than the pooled pattern predicts.

The plot establishes a descriptive association. Causal interpretation would require an identification strategy and evidence on competing explanations.

Logarithmic axes represent proportional differences

Equal ratios receive equal visual distances:

$$1{,}000 \to 2{,}000 \qquad \text{and} \qquad 50 \to 100.$$

- Income spans more than two orders of magnitude.
- Mortality ranges from roughly 1.4 to nearly 500 deaths per 1,000.
- The briefing concerns proportional differences in income and mortality.

Axis labels remain in the original units. The spacing is logarithmic.

Each economy contributes repeated observations

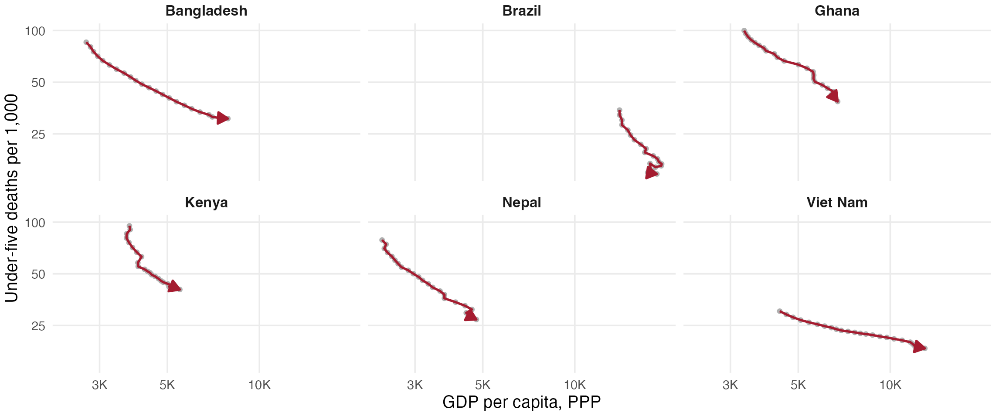
In the pooled plot, an economy can appear up to 23 times. The point cloud combines:

- persistent differences across economies;
- changes within economies over time;
- common movements between 2000 and 2022;
- modest differences in annual coverage.

The pooled association draws on both cross-sectional and longitudinal variation.

Before plotting the six trajectories, predict whether every selected economy will move along the pooled association.

Six economy trajectories show the longitudinal variation



Source: World Bank, World Development Indicators. Arrows follow each economy from earlier to later observed years.

The two figures use different comparisons

	Pooled plot	**Economy trajectories**
Unit shown	Economy–year	Repeated observations for one economy
Comparison	Higher-income and lower-income observations	Earlier and later observations within an economy
Use	Describe the overall association and dispersion	Document heterogeneity in changes over time
Limitation	Combines between- and within-economy variation	Selected economies do not summarize the full sample

The models will state which comparisons contribute to each coefficient.

Checkpoint 1: answer the director

A director sees the pooled plot and says: “Economic growth reduces child mortality. We should put that in the briefing.”

1. one claim the plot supports;
2. one reason the causal sentence exceeds the evidence;
3. one potential analysis we would like to implement.

2 minutes discuss $\longrightarrow$ **2 minutes share**

Checkpoint 1: model response

1. **Supported claim:** Across observed economy–years during 2000–2022, higher GDP per capita is associated with lower under-five mortality.
2. **Causal gap:** The specification does not isolate exogenous variation in income. Health systems, institutions, conflict, inequality, and other determinants can change with income.
3. **Next comparison (potential analysis):** Changes within the same economy after accounting for common year movements.

Ten-minute break

Return in ten minutes.

Leave the Lesson 3 walkthrough open at Section 4 in RStudio.

Lesson map

Capabilities for a changing labor market

Compare economies and changes over time

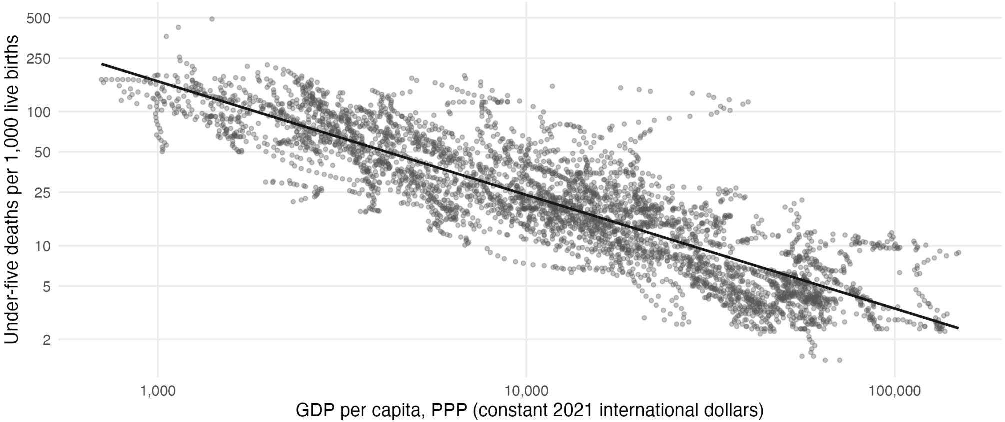
The specification determines the variation used to estimate the coefficient

Estimate, translate, compare

Check one briefing sentence with Codex

From lesson to lab

Pooled OLS summarizes the plotted association



Source: World Bank, World Development Indicators. The fitted line uses all observed economy–years and both variables in logarithms.

Fit the pooled association in RStudio

Predict the sign of `log_income`. Then run the model and record the coefficient and sample size.

```
fit_pooled <- lm(
  log_mortality ~ log_income,
  data = health_panel
)

summary(fit_pooled)
```

The model uses the same 4,296 observations shown in the plot.

Choose the correct interpretation

Slido multiple choice

Which sentence accurately reports the pooled estimate?

1. Across observed economy–years in 2000–2022, 10% higher GDP per capita is associated with about 7.8% lower under-five mortality on average.
2. Across observed economy–years, 10% higher GDP per capita is associated with 7.8 fewer deaths per 1,000 live births.
3. Within the same economy over time, 10% higher GDP per capita is associated with 7.8% lower mortality.
4. A 10% increase in GDP per capita reduces under-five mortality by 7.8%.

Join at sli.do

#2209653

Translate the log–log coefficient

The pooled coefficient on `log_income` is approximately -0.847.

For a 10% income difference:

$$100\left[\exp\{\widehat{\beta}\log(1.10)\} - 1\right] \approx -7.8\%.$$

Across observed economy–years during 2000–2022, 10% higher GDP per capita is associated with roughly 7.8% lower under-five mortality on average.

Predict the fixed-effects comparison

Slido multiple choice

After economy and year fixed effects are added, which variation identifies the income coefficient, and how do you expect its magnitude to compare with pooled OLS?

1. Within-economy changes after common year movements; smaller magnitude than pooled OLS.
2. Cross-economy comparisons within a calendar year; approximately the same magnitude as pooled OLS.
3. Changes in the global annual mean; larger magnitude than pooled OLS.
4. Economies observed in 2022; a coefficient of zero.

Join at sli.do
#2209653

Three specifications use different sources of variation

1. **Pooled OLS: comparisons across all economy–years.**
   Persistent economy differences and year movements contribute.
2. **Year fixed effects: economies compared within a calendar year.**
   Common year movements are absorbed. Persistent economy differences remain.
3. **Economy and year fixed effects: changes within economies.**
   Economy levels and year movements are absorbed. Other covariates may still change.

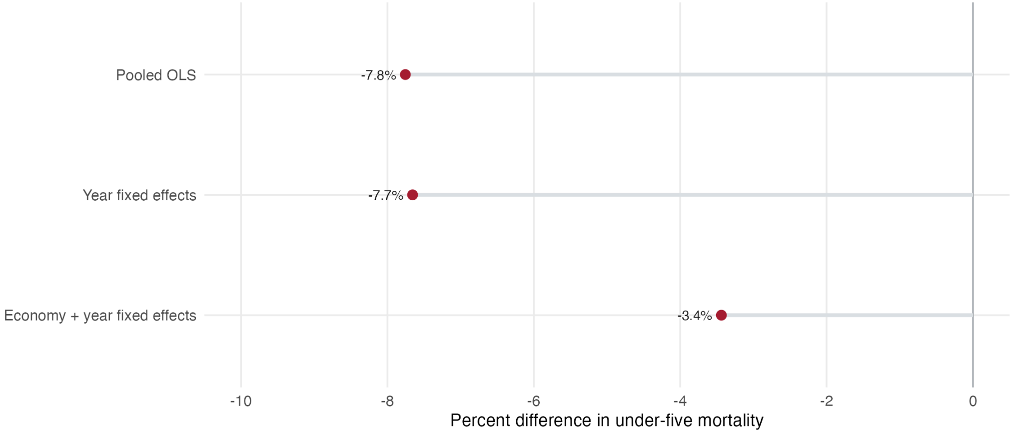
Each specification is descriptive. The variation used to estimate the income coefficient changes.

Change the specification in RStudio

```
fit_year_fe <- lm(
  log_mortality ~ log_income + factor(year),
  data = health_panel
)

fit_economy_year_fe <- lm(
  log_mortality ~ log_income +
    factor(iso3c) + factor(year),
  data = health_panel
)
```

The within-economy estimate is smaller



Source: authors' calculations using World Bank World Development Indicators. Each estimate reports the mortality difference associated with 10% higher GDP per capita.

Interpret the three estimates

- **Pooled OLS:** about -7.8% across all economy–years.
- **Year fixed effects:** about -7.7% across economies observed in the same calendar year.
- **Economy and year fixed effects:** about -3.4% from changes within economies after common year movements.

Year fixed effects barely change the pooled estimate. Adding economy fixed effects reduces the coefficient substantially. Persistent cross-economy differences account for most of the change across specifications.

The figure reports point estimates. Inference requires standard errors that allow for dependence within economies.

Checkpoint 2: draft a three-sentence briefing

With a partner, write exactly three sentences:

1. Report the pooled estimate in percentage units.
2. Report how the economy and year fixed-effects estimate differs and state which comparisons changed.
3. State the principal research-design limitation.

2 minutes discuss $\longrightarrow$ **2 minutes share**

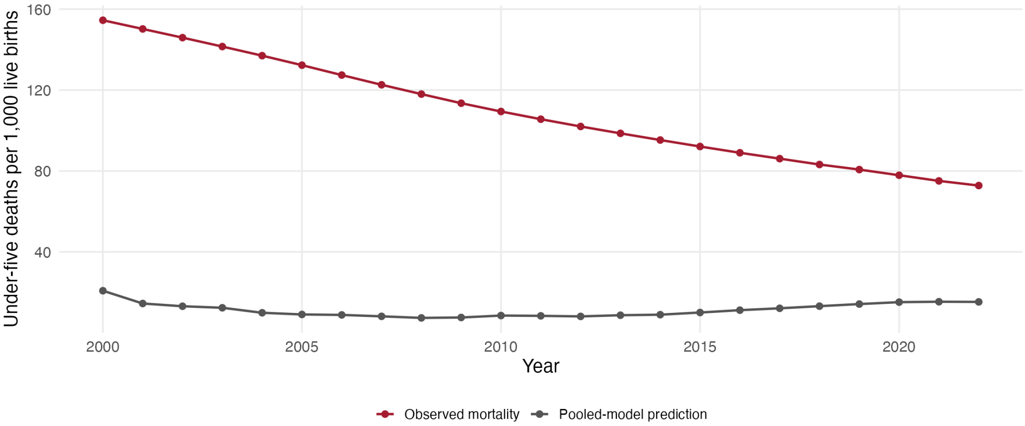
Checkpoint 2: model briefing

Across 4,296 observed economy–years during 2000–2022, 10% higher GDP per capita is associated with about 7.8% lower under-five mortality on average.

With economy and year fixed effects, the corresponding association is about 3.4%; this coefficient uses changes within economies after accounting for common year movements.

Other time-varying determinants remain, so these estimates do not identify the effect of an income intervention.

Equatorial Guinea departs sharply from the pooled prediction



Source: World Bank, World Development Indicators, and authors' calculations. The gray series applies the pooled log–log specification to Equatorial Guinea's reported GDP per capita in each year.

The country case raises additional policy questions

In 2008, Equatorial Guinea's reported GDP per capita was approximately $39,995 in constant PPP terms. Under-five mortality was 118 deaths per 1,000 live births; the pooled model predicted approximately 7.4.

Before explaining the discrepancy, investigate:

1. the definitions, units, and construction of both indicators;
2. how national income was distributed across households;
3. access to primary care, sanitation, nutrition, and maternal services;
4. the timing between changes in income and changes in health systems;
5. whether the pooled functional form represents this country's trajectory.

The figure establishes a large difference between the observed outcome and the pooled prediction. Explaining that difference requires additional data.

Does one observation drive the pooled estimate?

Remove Equatorial Guinea–2008, refit the same pooled specification, and compare the reported 10% income difference.

Estimation sample	Observations	Estimated difference
All economy–years	4,296	-7.755%
Excluding Equatorial Guinea–2008	4,295	-7.760%

Removing the observation barely changes the pooled coefficient. This check addresses the sensitivity of that coefficient to one observation. It does not explain Equatorial Guinea's health trajectory.

Lesson map

Capabilities for a changing labor market

Compare economies and changes over time

The specification determines the variation used to estimate the coefficient

Check one briefing sentence with Codex

Specify, reproduce, decide

From lesson to lab

Specify the claim check

Element	Lesson 3 claim check
Task	Check one reported sentence against the saved model results.
Files	`model-comparison.csv`, the checked analysis table, and the R code.
Permission	Read files and run checks; do not edit project files.
Verification standard	Match the number to its specification and independently recompute one result.

The task, files, permissions, and verification standard are fixed before the agent starts.

The audit checks four components of the empirical analysis

The project skill `policy-claim-audit` asks the agent to:

1. match the policy question to the unit, sample, and period;
2. verify the figure's variables, scales, source, and observations;
3. match each coefficient to its specification and identifying variation;
4. separate supported claims, empirical concerns, and hypotheses.

The response must cite named files or R objects and report an independent check.

Ask Codex to check the briefing sentence

```
Use $policy-claim-audit to check this sentence:
"After economy and year fixed effects are included, 10%
higher income is associated with about 3.4% lower mortality
within economies. Excluding Equatorial Guinea-2008 barely
changes this estimate. This shows that raising income reduces
under-five mortality."

Read code/lesson-3-walkthrough.R,
outputs/lesson-3/model-comparison.csv, and
outputs/lesson-3/sensitivity-record.csv. Do not edit files or
browse the web. Recompute both reported comparisons. Identify
the specification behind each one. Assess the causal language,
cite the relevant file or R object, and propose corrected text.
```

Checkpoint 3: evaluate the agent's claim

The agent reports:

> *"After economy and year fixed effects are included, 10% higher income is associated with about 3.4% lower mortality within economies. Excluding Equatorial Guinea–2008 barely changes this estimate. This proves that income growth lowers mortality."*

1. underline the supported empirical statement;
2. identify the specification behind each numerical statement;
3. identify the specification mismatch and unsupported inference;
4. rewrite the briefing sentence.

Lesson map

Capabilities for a changing labor market

Compare economies and changes over time

The specification determines the variation used to estimate the coefficient

Check one briefing sentence with Codex

From lesson to lab

Carry the same analysis forward

Re-rank the five capabilities

Slido ranking

A coding agent produced a figure and paragraph for a ministerial briefing due in 30 minutes. Rank the same five capabilities again:

1. Frame a policy question with a decision at stake.
2. Exercise statistical judgment.
3. Steward data and provenance.
4. Read, modify, and explain code.
5. Supervise and verify an agent.

Compare this ranking with your opening response. Which analytical move changed your ordering?

Join at sli.do

#2209653

Lab 3 has four exercises before Codex

During the next hour, each student will:

1. select and describe one documented region;
2. construct parallel all-countries and regional briefing tables;
3. construct comparable all-countries and regional figures;
4. write an interpretation and save the R file.

After Exercise 4, Codex reads the same saved file, fits one pooled model for each sample, and proposes claims. You verify one material statement before deciding whether to accept, revise, or reject it.

A policy briefing has five required components

Question	Sample	Figure	Model	Sentence
Audience and task	Unit, key, period	Pattern and dispersion	Comparison used	Estimate and limitation

Open `code/lab-3-starter.R`. Keep the selected region consistent across the briefing table, figure, model request, and final sentence.

A policy claim is ready for use when the question, sample, figure, specification, and sentence refer to the same comparison.

References

Green, Andrew, “Artificial Intelligence and the Changing Demand for Skills in the Labour Market,” OECD Artificial Intelligence Papers 14, OECD Publishing, Paris 2024.

OECD, “Skills in the AI Age,” OECD Artificial Intelligence Papers 60, OECD Publishing, Paris 2026.

Pocock, Matt, “AI Coding Dictionary,” AI Hero 2026. Entries consulted: Model, Effort, Agent, Harness, Tool, Tool Result, Model Provider Request, and Skill.

World Economic Forum, “The Future of Jobs Report 2025,” Technical Report, World Economic Forum, Geneva 2025.